

MATH 350-2 Advanced Calculus

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September 2, 2025

Outline

- 1 Real Analysis Lecture 5
 - Set basics
 - Set algebra

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- $\{n \in \mathbb{Z} : n \text{ is prime}\}$

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- even relations and functions are sets!

Challenge!

Problem

Prove that for ordered pairs (a, b) and (c, d) that

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Case I: $a = c$ and $\{a, b\} = \{c, d\}$

Case II: $a = \{c, d\}$ and $\{a, b\} = c$

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Therefore $b = c$ or $b = d$.

If $b = d$, we're done!

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Then $a = c$ implies $a = b$.

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It follows that $d = c = b = a$.

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This can be shown to contradict the ZF Axioms of Set Theory.

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This can be shown to contradict the ZF Axioms of Set Theory.
Specifically the regularity axiom for the set $\{a, c\}$...

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NOTATION:

$$\bigcup_{i=1}^n A_i = A_1 \cup A_2 \cup \cdots \cup A_n.$$

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Challenge

- index set $I = (1, 2)$
- family of sets $\{A_i : i \in I\}$
- $A_i = [0, i]$

Problem

Determine $\bigcup_{i \in I} A_i$.

Challenge

- index set $I = \mathbb{Z}_+$
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- $A_i = [0, i)$

Problem

Determine $\bigcap_{i \in I} A_i$.

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